

Forecasting Epidemics

Forecasting and Nowcasting with Text as Data

Homework 1

Members: Èric Gutiérrez, Gal·la Gelpí, Tizian Schenk

May 23, 2026

1 Introduction

In 2023, the world registered 8.22 million deaths from infectious diseases (IHME, 2025), as well as several billions of cases. Moreover, and between 1996-2023, a total of 3,013 global outbreak events were reported (Liu et al., 2025). The effects of this epidemic outbreaks for nations worldwide are devastating, and transcend health. Firstly, the health and well-being of the population in the affected areas is endangered, and secondly, national economies are impacted and development is halted, as resources are diverted towards fighting disease (Fan et al., 2018; Jones et al., 2008). In addition, many of the nations with the highest frequency of outbreaks are in conflict-ridden and conflict-prone areas (Liu et al., 2025); with epidemic outbreaks imposing additional challenges to first responders and causing even more harm to an already suffering population.

In the light of the riveting effects that infectious diseases have worldwide, and as Liu et al. (2025) conclude (Liu et al., 2025), there is a need for early detection and rapid response systems. In this paper, we present a text-enriched model to forecast epidemic outbreaks at the national, monthly level. We frame epidemic outbreaks as a prediction policy problem (Kleinberg et al., 2015), where a policy's effectiveness is given both by the causal effect of the policy and by a targeting exercise. Specifically, the model we present aims at providing a solution for the latter. In other terms, this paper does not explore the causal effects of policies, nor does it investigate which policies should be implemented, but rather tries to answer "where" an epidemic outbreak will happen in the coming months.

Mueller and Rauh (2022) define the hard conflict of conflict prediction as the prediction of outbreaks of violence in countries without a recent history of violence. Similarly, we define the hard conflict of epidemic forecasting as the prediction of outbreaks in countries without a recent history of infectious diseases epidemics. By introducing this notion, we increase the complexity of the forecasting exercise, as we are now also interested in highly unlikely events; which are harder, but arguably more interesting to study.

The outline of this paper is the following: first, this section introduces and motivates the topic of epidemic outbreaks. In section 2, we detail which data has been used for the forecasting exercise, which is presented in depth in section 3. In section 4, we provide our model's results, as well as a cost model to justify all the decision taken in our classification strategy. Finally, section 5 concludes.

2 Data

The analysis draws on four data sources merged on a common country-month index. The resulting panel covers 173 countries from January 2010 through December 2024, yielding 31,122 observations. The final 24 months (January 2023 through December 2024) are reserved for pseudo-out-of-sample evaluation, producing 4,152 test observations of which 122 (2.94 percent) are positive outcomes.

2.1 Target Variable: EM-DAT Epidemic Events

The outcome variable is constructed from the Emergency Events Database (EM-DAT), maintained by the Centre for Research on the Epidemiology of Disasters (CRED) at UCLouvain (Centre for Research on the Epidemiology of Disasters, 2024). We retain only events classified as epidemics, excluding insect and worm infestations. The binary indicator $\texttt{epidemy}_{it}$ equals one if at least one epidemic began in country i during month t and zero otherwise. A balanced panel is constructed for all 131 countries with at least one recorded epidemic, spanning January 2000 through December 2025 (312 months per unit, 40,872 country-month observations in total).

Following Mueller et al. (2024) and Kleinberg et al. (2015), we construct a forward-looking incidence target at horizon $h = 4$ months:

$$\texttt{target}_{it}^{(4)} = \max\big(\texttt{epidemy}_{i,t+1},\, \texttt{epidemy}_{i,t+2},\, \texttt{epidemy}_{i,t+3},\, \texttt{epidemy}_{i,t+4}\big)\,, \tag{1}$$

which equals one if any epidemic begins within the next four months and zero otherwise. The four-month horizon matches the logistical lead time required to pre-position WHO emergency response resources (Fan et al., 2018).

2.2 Structural Health Indicators: World Bank

Annual health indicators are drawn from the Health, Nutrition and Population Statistics collection of the World Bank World Development Indicators (World Bank, 2024). Starting from 66 annual indicators, we drop five series with no observations prior to 2000, retaining 61 covering health expenditure, immunisation coverage, water and sanitation access, disease incidence, and healthcare capacity for the period 2000–2024. Annual values are lagged by one year and interpolated to monthly frequency by distributing the year-on-year change uniformly across the following twelve months, eliminating look-ahead bias by construction. Lags at 3, 6, and 12 months and rolling means and standard deviations over 3 and 6 months are computed for each raw indicator, producing 366 derived World Bank features. A pre-selection step using a Random Forest trained on data prior to January 2023 retains the 8 most informative raw and the 8 most informative derived World Bank features.

2.3 Text Features: Conflict Forecasting News Topics

Text-based features are drawn from the monthly news topic stocks published at https://conflictforecast.org by Mueller et al. (2024). A dynamic Latent Dirichlet Allocation model with 15 topics (Blei et al., 2003) is estimated on a corpus of over six million newspaper articles. Country-month topic shares are transformed into exponentially weighted moving averages with decay $\delta = 0.8$, giving more weight to recent news (Mueller et al., 2024). Data cover 174 countries from January 2010 through December 2024.

The 15 topics are listed in Table 1. For each raw topic stock, we additionally compute lags at 1, 2, and 3 months, exponentially weighted rolling means over 3 and 6 months, rolling standard deviations over 3 and 6 months, and rolling maxima over 3 months, yielding 120 derived text features.

Pre-selection retains the 10 most informative derived text features. In the best-performing model specification, derived features from the *Diplomacy* topic (topic 11) account for 67.1 percent of total feature importance and *Military Conflict* (topic 2) for 3.6 percent, consistent with the hypothesis that international diplomatic engagement and active conflict serve as upstream proxies for the state fragility that precedes epidemic outbreaks (Connolly et al., 2004; Jones et al., 2008). For the 19 countries absent from the news corpus, topic stock values are set to zero.

Table 1: LDA news topic labels (Mueller, Rauh & Seimon, 2024, Table 6) and epidemic relevance assessed via the epidemiological instability mechanism.

Index	**Label**	**Epidemic relevance**
0	Competition and Sports	None
1	Health and Education	Medium
2	Military Conflict	High
3	Politics	Low
4	Military Technology	Low
5	National Development	Low
6	Political Tensions	Medium
7	Judiciary and Abuses	Medium
8	Middle East	Medium
9	Chinese Politics	Low
10	Economics	Low
11	**Diplomacy**	**Highest**
12	Civilian Life	High
13	Foreign Policy	Low
14	Power and Negotiation	Low

2.4 Anticipatory Surveillance Signal: WHO Disease Outbreak News

The WHO Disease Outbreak News (DON) repository, accessed via the Humanitarian Data Exchange (World Health Organization, 2024), records outbreak events from 1998 onwards across 285 country units under the International Health Regulations framework. COVID-19 entries are excluded as they represent sustained administrative reporting rather than discrete onset events. The remaining records are aggregated to an annual country-level count and assigned to all twelve months of the following calendar year ($\texttt{alertes_count}_{it}$ for country i in year Y equals the DON report count for year $Y-1$), precluding look-ahead bias by construction.

2.5 Final Panel

The four sources are merged on a common `(ISO, month)` multi-index. The model panel is the intersection of country-month cells present in both the World Bank regressors and the text topic data, yielding 173 countries and 180 monthly periods from January 2010 through December 2024. The restriction to January 2010 is imposed by the earliest available topic stock data; the World Bank and WHO DON series extend back to 2000 and 1998, respectively, allowing epidemic history features to draw on up to ten years of prior activity at the start of the model window.

3 Methodology

In this section, we present the methodology that we have followed to build our text-enriched model to forecast epidemic outbreaks at the national level, with a monthly periodicity, and within a four-month window. In other terms, the model's predictions correpond to the probability that a given country experiences an epidemic outbreak in the next four months. This particular horizon was selected so that a balance between predictive power, which is achieved by increasing the horizon, and relevance, which benefits from its reduction, is achieved.

The first step involves data preparation, which corresponds to formatting the data sources that were presented in the previous section for a posterior feature engineering stage. The target variable, the epidemic outbreaks from the EM-DAT, has been expanded so that all of the 173 countries in the dataset have a value for each of the months for the 2010-2024 period, populating all the previously missing rows with zeros.

Two distinct operations were performed on the data from the Health World Bank. Firstly, and with the purpose of dealing with missing data, we opted for propagating forward in time the existing data. For example, if the years 2003 and 2004 had no data, we took the value from the latest year with data available before 2003, that is, 2002. Notice that by choosing forward propagation, as opposed to backward propagation, we avoid data leakage by ensuring that for each month, the model only uses the information that was necessarily known up to that point.

Secondly, the Health World Bank data is reported yearly, but the frequency of our model is monthly. In order to bring the data to our desired level of temporal disaggregation, we had to come up with a solution that approximated the changes in the health indicators at the month level, while strictly avoiding data leakage. We settled for shifting the yearly data one year, with the purpose of ensuring that the information that would have been available at the beginning of that year was the only information ingested by the model. Finally, we interpolated each month's value by progressively applying a monthly trend derived from historical year-over-year changes.

The anticipatory surveillance signal, the WHO Disease Outbreak News, was also expanded so that all combinations of countries in the dataset and months in the period of interest are included. As for the conflict forecasting news topics, no data preparation step was required.

Moreover, we transformed the target variable into an incidence variable. We have chosen to forecast the incidence and not the onset, for instance, because we are interested in every one of the epidemic outbreaks, and thus, we do not want to discard data corresponding to an outbreak that immediately follows another one. Consequently, the target variable is 1 if an epidemic outbreak was registered in the following four months, and 0 otherwise.

After the data preparation stage, we turn to feature engineering to further enrich our model. For each one of the health indicators provided by the Health World Bank, we included a 3-month, a 6-month, and a 12-month lag, as well as the rolling mean with a 3-month and a 6-month window, and a 6-month rolling standard deviation. As for the topic stocks from the conflict forecasting news topics, we introduced to the model lags with windows 1, 2, and 3 months, in addition to two weighted rolling means (with windows 3 and 6, and a decay factor of 0.8), two rolling standard deviations (with windows 3 and 6), and a 3-month rolling maximum.

For the target variable, we engineered a since feature, that counts the number of months since the last epidemic outbreak for a given country. Several rolling means with 1-, 3-, 6-, 12-, 15-, 21-, and 24-month windows were also included. Notice that the selection of this specific features is not fortuitous. In order to only include the best-performing features and potentially overcome the curse of dimensionality, we employed a preliminary Random Forest Classifier, from which we computed the feature importance across the window space. The features detailed above, therefore, are those that were found to be more relevant for predicting epidemic outbreaks.

We implement a rolling forecast with rolling pseudo-out-of-sample validation. In other words, we run a chronological backtest to simulate how the model would have performed month by month over the last 2 years. We use PanelSplit with the following specifications: first, we generate 24 sequential test folds, each corresponding to one months (covering the two-year test period from January 2023 to December 2024). Second, we set the test size as 1, as each fold predicts exactly one month ahead. The gap is set at 3, provided that we forecast with a four-month horizon, solving the data leakage between the test and the train.

We use a Random Forest Classifier with the hyperparameters facilitated by Ben Seimon, specifically: maximum features as 0.2, a maximum depth of 4, using balanced class weigths, and with a minimum number of samples required to be at a leaf node of 100. Importantly, we do not restrict our analysis to training only one model with all the data that we have prepared and engineered as described in the previous sections, but rather, we generate predictions with several models with different data inputs. In the next section, we present two of the models, namely: our baseline model with all the data retrieved expect for the text inputs, and a text-enriched model.

Furthermore, and in the ROC curve space resulting from the text-enriched model, we introduce economic modelling as a tool to justify the selection of the balance between any given policy's intervention cost and its effectiveness. This two components are the gears that set the slope of the isocost lines in the ROC curve space, as well as the optimal threshold c for the classification model (Mueller and Rauh, 2022). Specifically, the slope of the isocost curves is given by:

$$\frac{\partial TPR}{\partial FPR} = \frac{Cost_{FP} - Cost_{TN}}{Cost_{FN} - Cost_{TP}} \frac{N}{P}$$

A False Positive occurs when the model issues an alert for an epidemic that does not materialize, triggering an unnecessary field deployment of personnel and countermeasures normalized to a baseline unit cost of 1.0. Conversely, a False Negative represents a catastrophic, irreversible failure to alert on a true incoming outbreak, resulting in unmitigated transmission, sharp mortality spikes, long-term morbidity, and localized economic paralysis. To mathematically reflect this ethical and operational reality, correct classifications (True Negatives and True Positives) are normalized to a cost of 0 because upfront intervention expenditures are entirely offset by the value of lives saved, while a severe penalty scales $Cost_{FN}$ to 25.0.

Instead of relying on the standard 50% cutoff, our model evaluates 101 different probability thresholds against these specific asymmetric business costs. This optimization framework allows us to isolate the exact customized threshold for each model configuration that minimizes real-world operational costs and maximizes the preservation of human life. Subjecting each experiment to this cost-sensitive matrix ultimately provides an empirical metric to evaluate how integrating different data streams

systematically helps reduce institutional expenditures and deployment frictions.

4 Results

In this section, we present our models' results, as well as our interpretation of their performance.

We evaluate three distinct modeling configurations designed to capture separate vectors of predictive signals for epidemic forecasting: a structural Baseline model (relying on historical epidemiology, Health World Bank indicators, and their derivatives), a Text Only model (leveraging latent text topics, derived text trends, and official WHO Disease Outbreak News), and a combined Baseline + Text model. The foundational Baseline model leverages historical trends under the assumption that past outbreak patterns replicate, while the Text Only model isolates real-time communications to assess if public health reporting can anticipate formal structural updates. Across the entire evaluation panel, the Baseline model achieves the best performance, with an Area Under the Receiver Operating Characteristic (AUC-ROC) curve of 0.7732, closely matched by the integrated Baseline + Text model at 0.7711. The Text Only model operates at a lower overall baseline capacity of 0.7322 AUC-ROC, but it exhibits a superior overall Average Precision (AP) of 0.0842 compared to the structural baseline's 0.0732, indicating a cleaner precision-recall profile in high-confidence prediction bands.

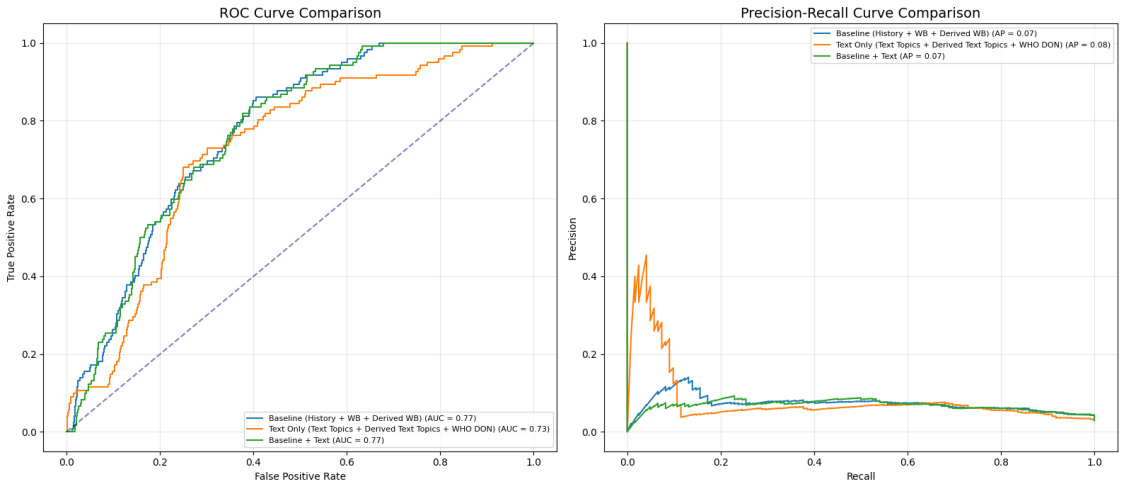


Figure 1: ROC and PR curve comparison for the general group of models.

To evaluate the specific predictive utility of text data, we stratify our panel data into soft incidence regimes (where an outbreak has occurred within the last 24 months) and hard incidence regimes (where a country has not registered any outbreak in over two years). In soft incidence settings, the Baseline model excels, achieving an AUC-ROC of 0.9068 because historical rolling means and recent infection tracking provide highly reliable autoregressive signals. However, when evaluating the hard onset problem where history is entirely silent, the predictive capacity of the structural baseline degrades sharply to an AUC-ROC of 0.6428. Under these historically silent conditions, the inclusion of text data shifts performance, causing the integrated Baseline + Text model to achieve a hard

AUC-ROC of 0.6665. This yields a net text-driven AUC-ROC gain of +0.0237 specifically when history is blind, confirming that unstructured informational features carry critical early-warning signals precisely when autoregressive indicators are absent.

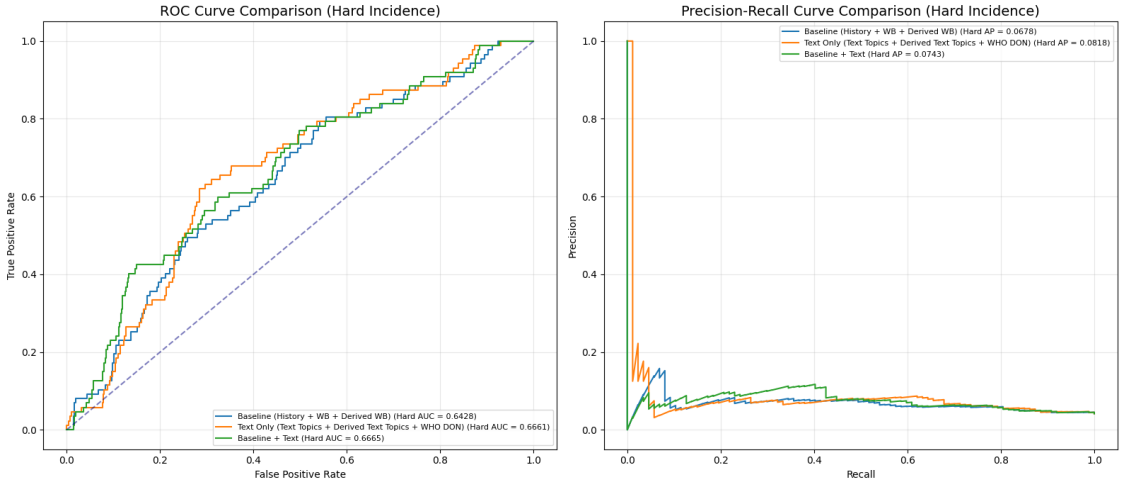


Figure 2: ROC and PR curve comparison for hard incidence.

The mechanics driving this performance split are revealed through the model's feature importance profiles across both regimes. In soft incidence cases, structural and historical tracking variables dominate the model's tree splits, with historical rolling means and the time elapsed since the last outbreak accounting for a combined 32.6 percent of total group importance. Conversely, in the hard onset regime, the importance of historical variables drops to a 1.1 percent, and the model structurally reallocates its split criteria toward unstructured data, with text topics capturing 35.5 percent of total feature group importance. When evaluating the Text Only configuration independently, structural text derivatives and raw textual indicators account for 72.3 percent and 26.4 percent of cumulative importance respectively. This clean structural shift demonstrates that the machine learning framework dynamically substitutes silent historical timelines with active global health reporting trends to generate predictive traction.

To translate these probabilistic outputs into actionable public health policies, we implement a cost-sensitive threshold optimization framework, as we introduced in the previous section. By assigning an asymmetric loss function that penalizes a missed epidemic twenty-five times more severely than a false alarm ($C_{FN}/C_{FP} = 25$), we calculate the optimal operating point on the empirical ROC curve using isocost line analysis. For the Baseline model, optimizing the decision boundary away from the naive 0.5 default to a tuned cutoff of 0.33 successfully recaptures a 4.1 percent reduction in total expected institutional and societal costs. The Text Only model maintains a steady default cost profile owing to regular probability calibration bounds across the full set, reinforcing the conclusion that combining text and structural indicators provides both statistical validation and stable economic utility for real-world deployments.

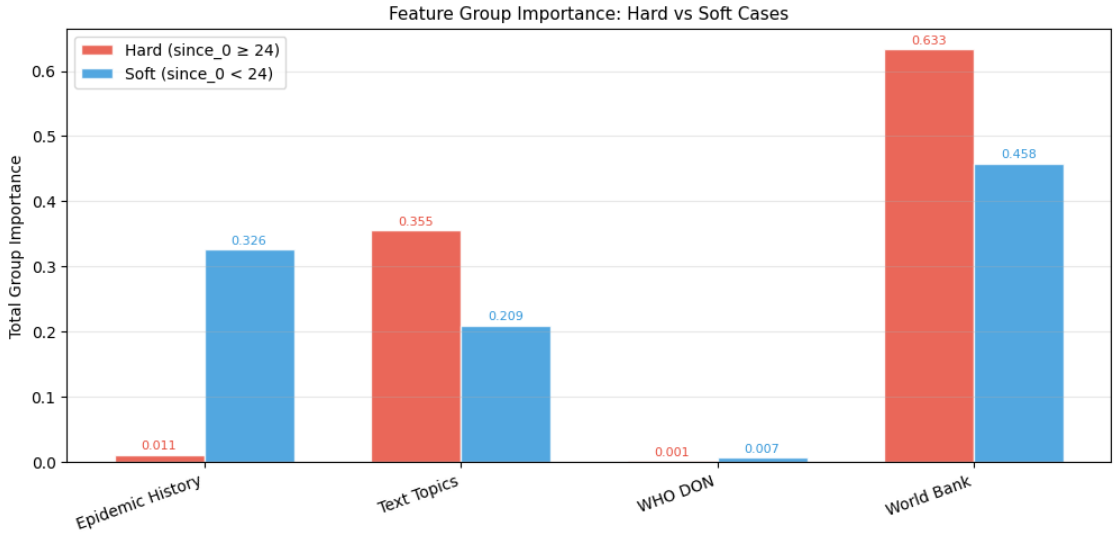


Figure 3: Feature importance, Hard vs Soft Incidence.

5 Conclusion

This paper developed a text-enriched early warning system to predict national epidemic outbreaks at a four-month horizon. We constructed a country-month panel covering 173 nations (2010-2024) by integrating World Bank structural health indicators, WHO Disease Outbreak News, and news topic stocks from a dynamic Latent Dirichlet Allocation model. Using a 24-month rolling pseudo-out-of-sample backtest, we evaluated a structural Baseline model against a text-enriched configuration to address the "hard prediction" problem of forecasting outbreaks amid extended epidemiological silence. While the Baseline excels in soft incidence settings with strong autoregressive signals (AUC-ROC of 0.9068), its performance collapses to 0.6428 under the hard onset regime. The integrated Baseline + Text configuration mitigates this blindness, generating a hard AUC-ROC gain of +0.0237 by allocating 35.5% of its feature splits toward open-source diplomatic and conflict news topics.

Nesting these findings within a cost-sensitive decision framework bridges the gap between statistical power and proactive policy. When penalizing a missed epidemic twenty-five times more severely than an unnecessary deployment, our optimized threshold framework yields a 4.1% reduction in expected institutional and societal costs. These savings validate our core premise: unstructured media text provides critical anticipatory indicators before formal, lagging public health frameworks update.

Nevertheless, key limitations present avenues for future research. These include reporting bias in the EM-DAT target variable that creates label noise for smaller outbreaks, a structural break from the 2020-2021 COVID-19 pandemic, and artificial temporal smoothness and publication lags inherent to annual World Bank interpolations. Finally, because the news corpus reflects conflict-focused tracking, sparse text coverage required zero-imputation for 19 country profiles—a data friction that future work can resolve using health-specific language model embeddings.

References

Blei, D. M., Ng, A. Y., and Jordan, M. I. (2003). Latent Dirichlet allocation. *Journal of Machine Learning Research*, 3:993–1022.

Centre for Research on the Epidemiology of Disasters (CRED) (2024). *EM-DAT: The Emergency Events Database* [Dataset]. UCLouvain, Brussels, Belgium. Available at: https://www.emdat.be.

Connolly, M. A., Gayer, M., Ryan, M. J., Salama, P., Spiegel, P., and Heymann, D. L. (2004). Communicable diseases in complex emergencies: impact and challenges. *The Lancet*, 364(9449):1974–1983. doi:10.1016/S0140-6736(04)17481-3.

Fan, V. Y., Jamison, D. T., and Summers, L. H. (2018). Pandemic risk: how large are the expected losses? *Bulletin of the World Health Organization*, 96(2):129–134. doi:10.2471/BLT.17.199588.

Jones, K. E., Patel, N. G., Levy, M. A., Storeygard, A., Balk, D., Gittleman, J. L., and Daszak, P. (2008). Global trends in emerging infectious diseases. *Nature*, 451(7181):990–993. doi:10.1038/nature06536.

Kleinberg, J., Ludwig, J., Mullainathan, S., and Obermeyer, Z. (2015). Prediction policy problems. *American Economic Review: Papers and Proceedings*, 105(5):491–495. doi:10.1257/aer.p20151023.

Liu, Q., Liu, M., Liang, W., Li, X., Jing, W., Chen, Z., and Liu, J. (2025). Global distribution and health impact of infectious disease outbreaks, 1996-2023: a worldwide retrospective analysis of World Health Organization emergency event reports. *Journal of Global Heatlh*, 15, 04151. Available at: https://doi.org/10.7189/jogh.15.04151.

Mueller, H. and Rauh, C. (2022). The hard problem of prediction for conflict prevention. *Journal of the European Economic Association*, 20(6):2440–2467. doi:10.1093/jeea/jvac025.

Mueller, H., Rauh, C., and Seimon, B. (2024). Introducing a global dataset on conflict forecasts and news topics. *Data & Policy*, 6:e17. doi:10.1017/dap.2024.10.

IHME, Global Burden of Disease (2025). Deaths from infectious diseases – IHME. *Global Burden of Disease – Deaths*, with major processing by Our World in Data. Retrieved May 22, 2026 from https://archive.ourworldindata.org/20260518-095641/grapher/deaths-from-infectious-diseases.html (archived on May 18, 2026).

World Health Organization (2024). *Global Pandemic and Epidemic-Prone Disease Outbreaks* [Dataset]. Humanitarian Data Exchange. Available at: https://data.humdata.org/dataset/global-pandemic-and-epidemic-outbreaks.

World Bank (2024). *Health Nutrition and Population Statistics* [Dataset]. World Development Indicators. Washington, D.C.: World Bank. Available at: https://databank.worldbank.org/source/health-nutrition-and-population-statistics.